

Analyzing Social Networks

Write a program in your preferred programming language that models a social network using an unweighted, undirected graph. In this graph, each node represents an individual user, and each edge represents a mutual friendship between two users. Your primary task is to implement the Breadth-First Search (BFS) algorithm to determine the shortest path between a given starting user and a target user. First, construct the graph using an adjacency list representation with a minimum of ten nodes. Then, execute your BFS traversal to find the minimum number of connections required to link the two specific individuals. Finally, output the exact sequence of users in the shortest path and the total degree of separation. Include brief, clear comments explaining the logic of your traversal, how the queue data structure is utilized, and how you keep track of visited nodes to prevent infinite loops during execution.

Analyzing Computer Networks

Write a script in your preferred programming language that models a computer network using an unweighted, undirected graph. In this graph, each node represents an individual router, and each edge represents a direct connection between two routers. Your primary task is to implement the Breadth-First Search (BFS) algorithm to determine the shortest path between a given starting router and a target router. First, construct the graph using an adjacency list representation with a minimum of ten nodes. Then, execute your BFS traversal to find the minimum number of hops required to link the two specific devices. Finally, output the exact sequence of routers in the shortest path and the total hop count. Include brief, clear comments explaining the logic of your traversal, how the queue data structure is utilized, and how you keep track of visited nodes to prevent infinite loops during execution.